Solution: Minimum Coin Change

Introduction

Given an infinite supply of ‘n’ coin denominations and a total money amount, we are asked to find the minimum number of coins needed to make up that amount.

Example 1:

```
Denominations: {1,2,3}
Total amount: 5
Output: 2
Explanation: We need a minimum of two coins {2,3} to make a total of '5'
```

Example 2:

```
Denominations: {1,2,3}
Total amount: 11
Output: 4
Explanation: We need a minimum of four coins {2,3,3,3} to make a total
of '11'
```

Problem Statement

Given a number array to represent different coin denominations and a total amount 'T', we need to find the minimum number of coins needed to make a change for 'T'. We can assume an infinite supply of coins, therefore, each coin can be chosen multiple times.

This problem follows the `"Unbounded Knapsack"` pattern.

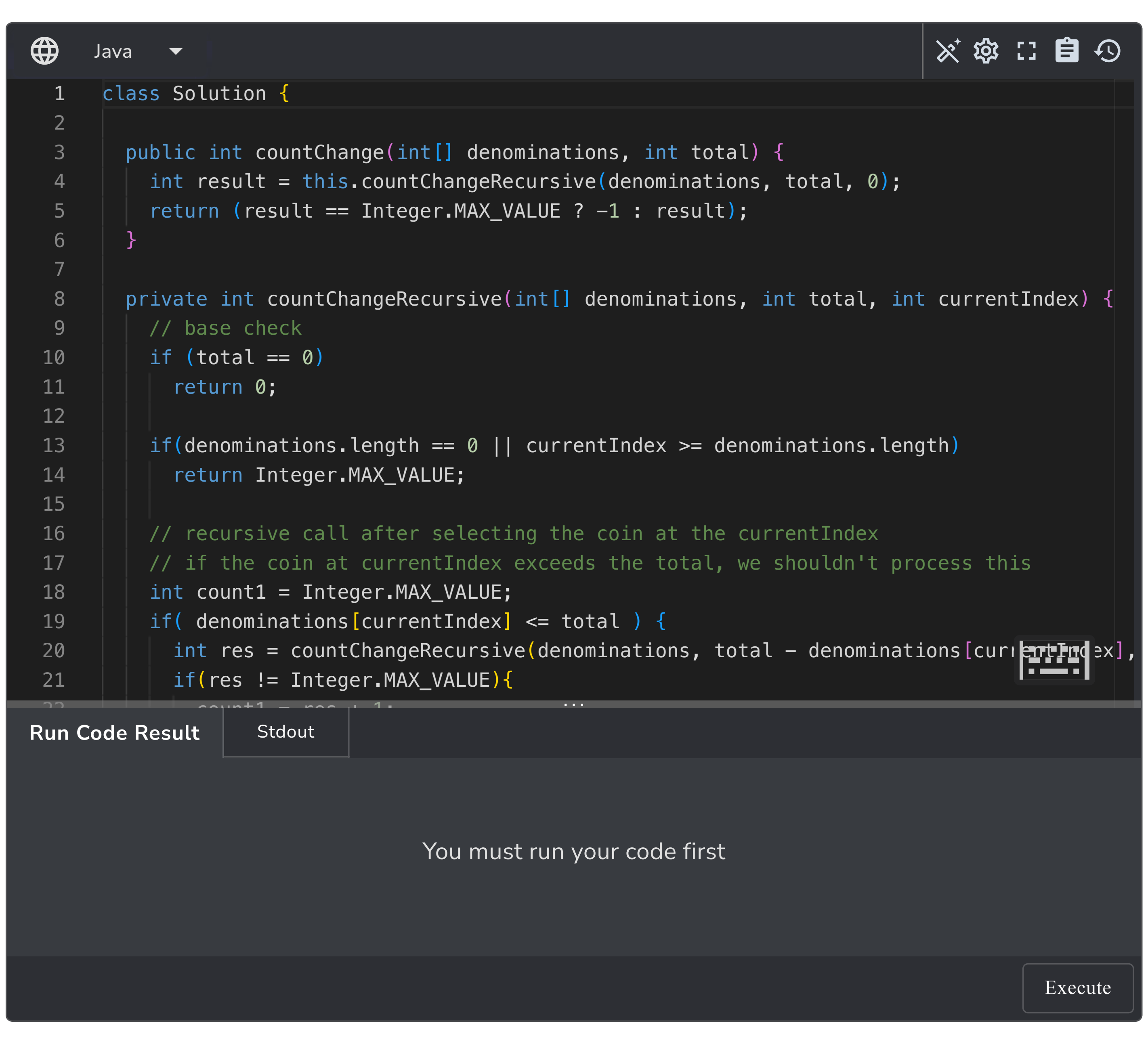
Basic Solution

A basic brute-force solution could be to try all combinations of the given coins to select the ones that give a total sum of 'T'. This is what our algorithm will look like:

```
for each coin 'c'
  create a new set which includes one quantity of coin 'c' if it does not exceed 'T',
and
     recursively call to process all coins
  create a new set without coin 'c', and recursively call to process the remaining coins
return the count of coins from the above two sets with a smaller number of coins
```

Code

Here is the code for the brute-force solution:



```java
class Solution {

  public int countChange(int[] denominations, int total) {
    int result = this.countChangeRecursive(denominations, total, 0);
    return (result == Integer.MAX_VALUE ? -1 : result);
  }

  private int countChangeRecursive(int[] denominations, int total, int currentIndex) {
    // base check
    if (total == 0)
      return 0;

    if(denominations.length == 0 || currentIndex >= denominations.length)
      return Integer.MAX_VALUE;

    // recursive call after selecting the coin at the currentIndex
    // if the coin at currentIndex exceeds the total, we shouldn't process this
    int count1 = Integer.MAX_VALUE;
    if( denominations[currentIndex] <= total ) {
      int res = countChangeRecursive(denominations, total - denominations[currentIndex],
      if(res != Integer.MAX_VALUE){
```

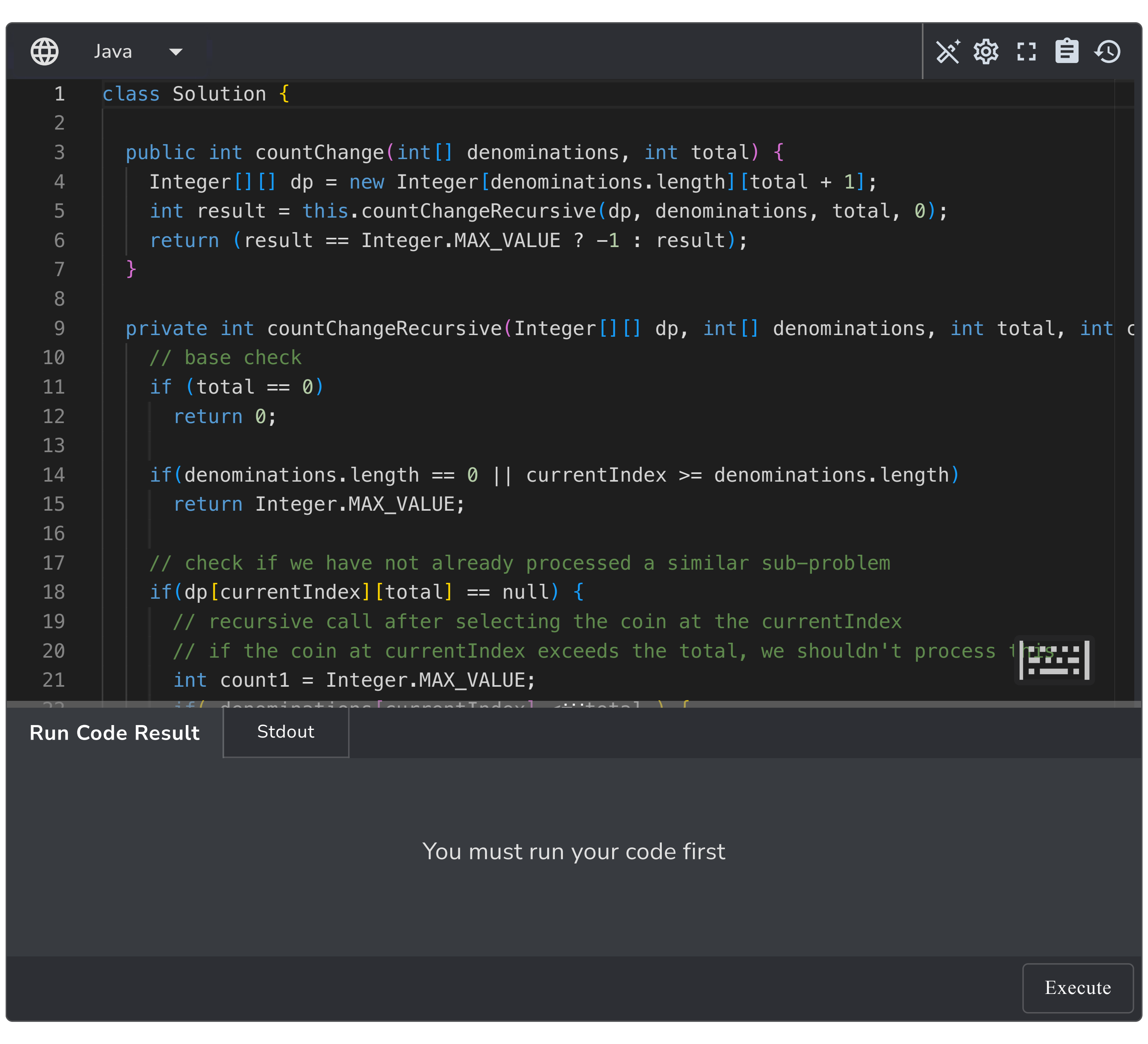
You must run your code first

The time complexity of the above algorithm is exponential $O(2^{C+T})$, where 'C' represents total coin denominations and 'T' is the total amount that we want to make change. The space complexity will be $O(C+T)$.

Let's try to find a better solution.

Top-down Dynamic Programming with Memoization

We can use memoization to overcome the overlapping sub-problems. We will be using a two-dimensional array to store the results of solved sub-problems. As mentioned above, we need to store results for every coin combination and for every possible sum:



```java
class Solution {

  public int countChange(int[] denominations, int total) {
    Integer[][] dp = new Integer[denominations.length][total + 1];
    int result = this.countChangeRecursive(dp, denominations, total, 0);
    return (result == Integer.MAX_VALUE ? -1 : result);
  }

  private int countChangeRecursive(Integer[][] dp, int[] denominations, int total, int c
    // base check
    if (total == 0)
      return 0;

    if(denominations.length == 0 || currentIndex >= denominations.length)
      return Integer.MAX_VALUE;

    // check if we have not already processed a similar sub-problem
    if(dp[currentIndex][total] == null) {
      // recursive call after selecting the coin at the currentIndex
      // if the coin at currentIndex exceeds the total, we shouldn't process this
      int count1 = Integer.MAX_VALUE;
```

You must run your code first

Bottom-up Dynamic Programming

Let's try to populate our array `dp[TotalDenominations][Total+1]` for every possible total with a minimum number of coins needed.

So for every possible total ‘t’ (0<= t <= Total) and for every possible coin index (0 <= index < denominations.length), we have two options:

- Exclude the coin: In this case, we will take the minimum coin count from the previous set => `dp[index-1][t]`
- Include the coin if its value is not more than ‘t’: In this case, we will take the minimum count needed to get the remaining total, plus include '1' for the current coin => `dp[index][t-denominations[index]] + 1`

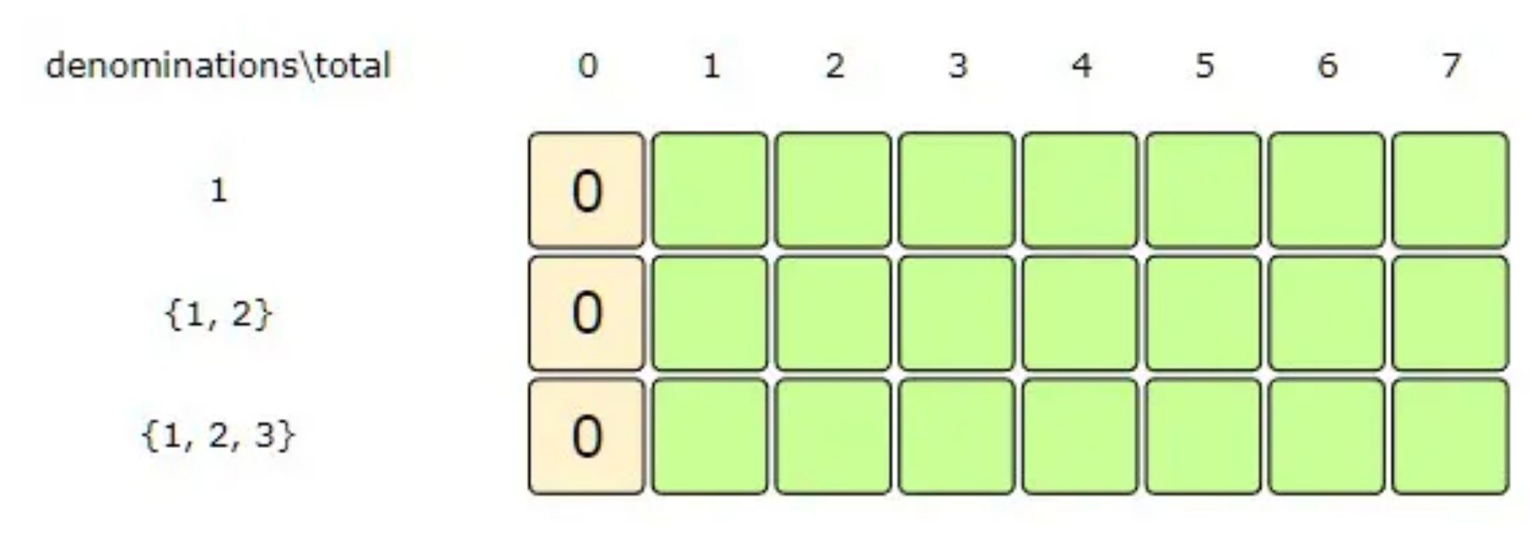
Finally, we will take the minimum of the above two values for our solution:

```
dp[index][t] = min(dp[index-1][t], dp[index][t-denominations[index]] + 1)
```

Let’s draw this visually with the following example:

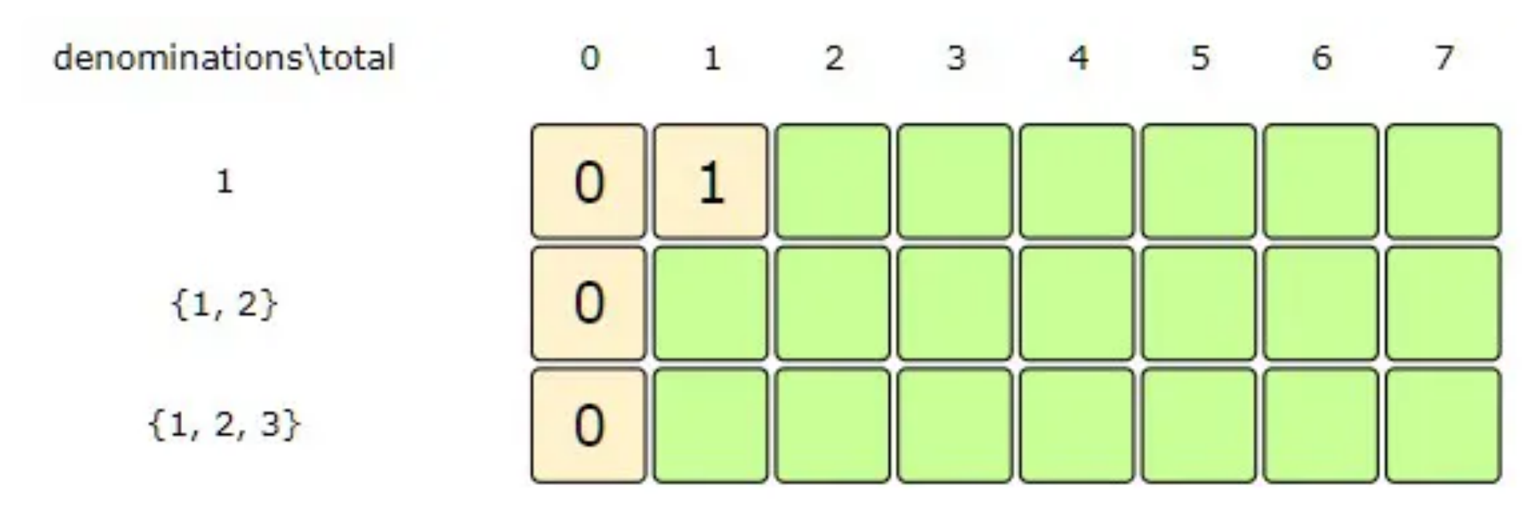
```
    Denominations: [1, 2, 3]
    Total: 7
```

Let’s start with our base case of zero total:



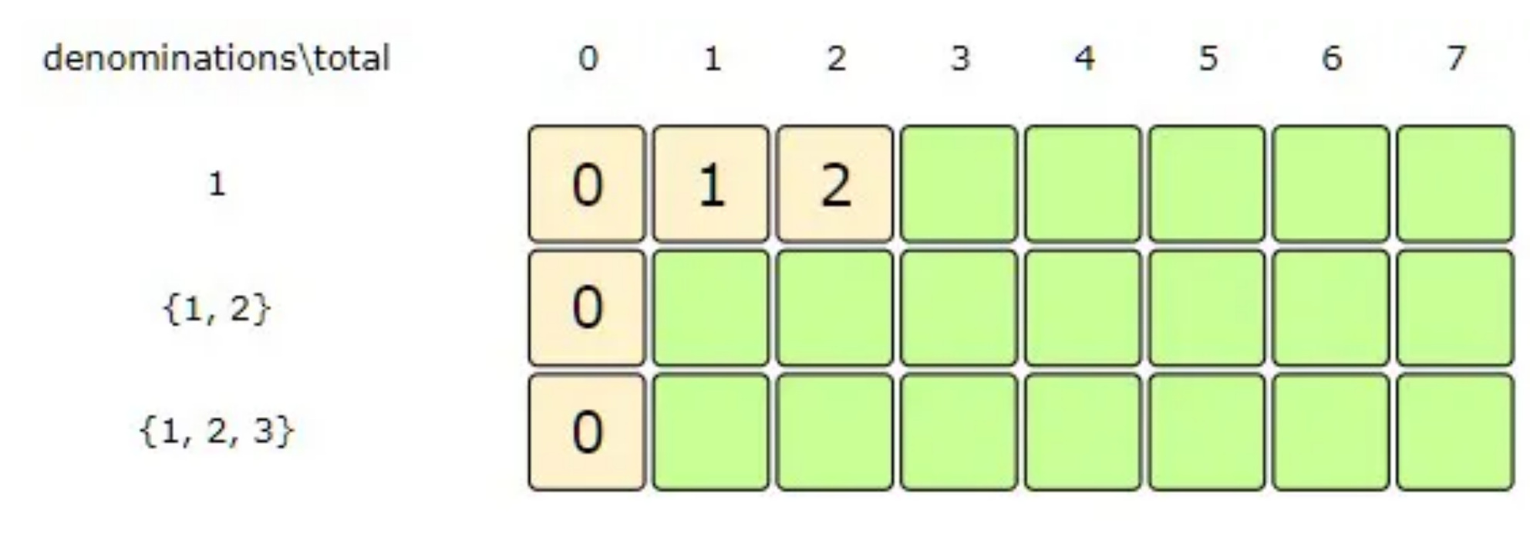
denominations\total	0	1	2	3	4	5	6	7
1	0							
{1, 2}	0							
{1, 2, 3}	0							

We don't need any coin to make zero total



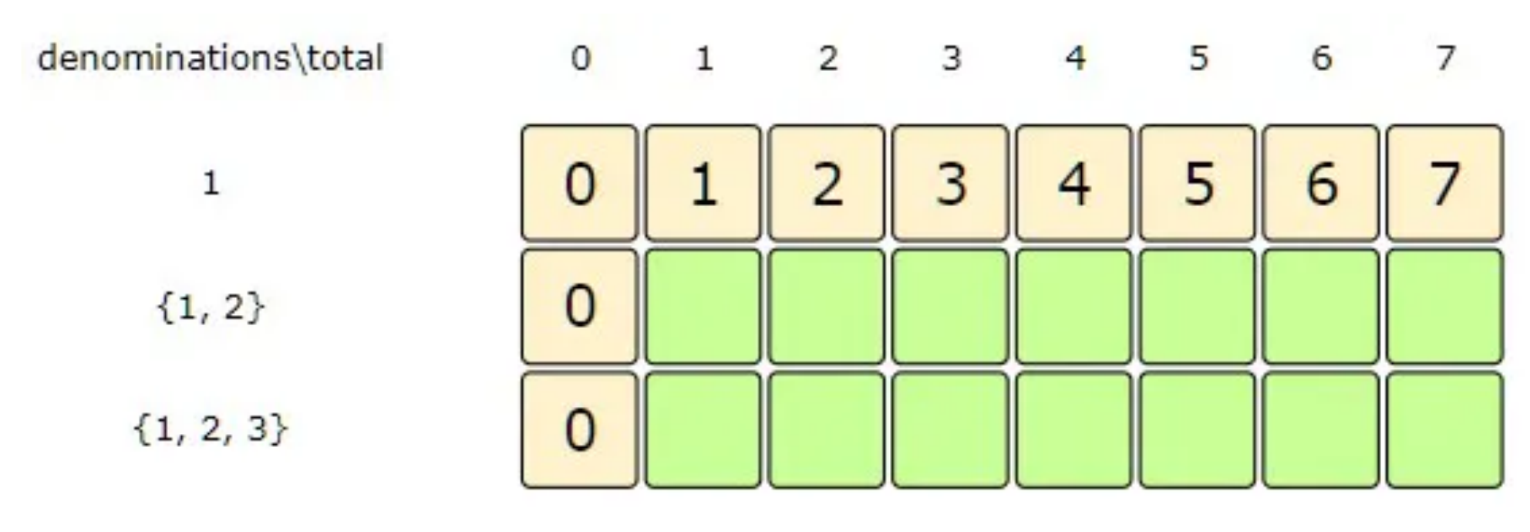
denominations\total	0	1	2	3	4	5	6	7
1	0	1						
{1, 2}	0							
{1, 2, 3}	0							

Total:1, Index:0 => dp[Index][Total-denominations[Index] + 1, we didn't consider dp[Index-1][Total] as Index = 0



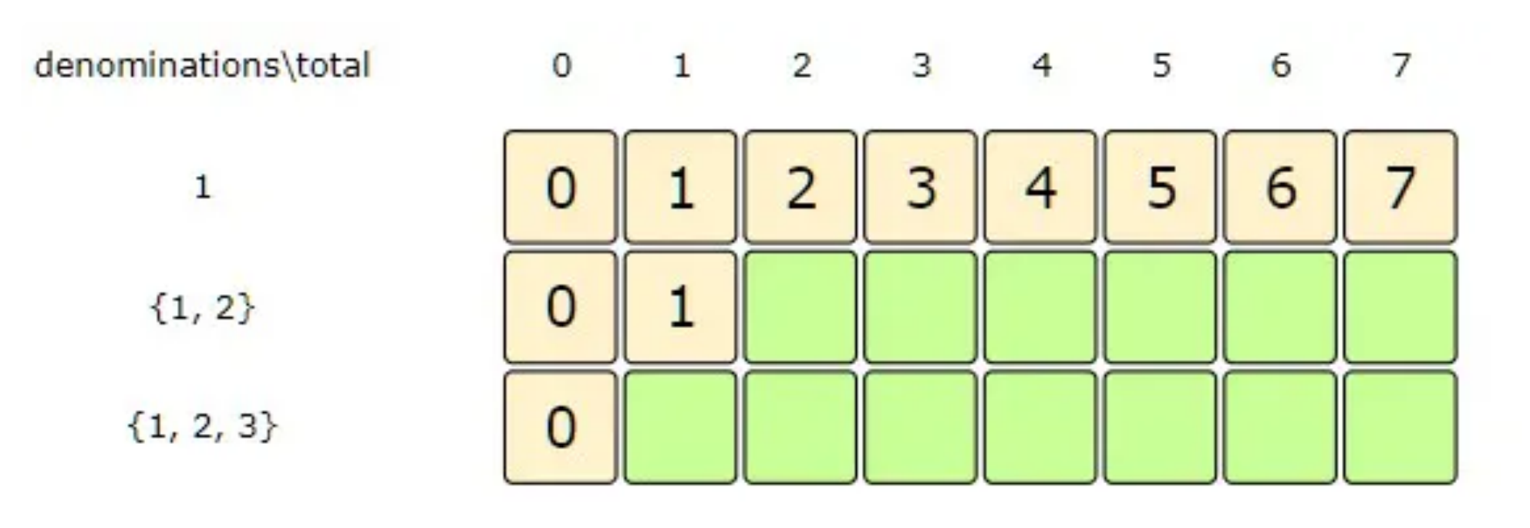
denominations\total	0	1	2	3	4	5	6	7
1	0	1	2					
{1, 2}	0							
{1, 2, 3}	0							

Total:2, Index:0 => dp[Index][Total-denominations[Index] + 1, we didn't consider dp[Index-1][Total] as Index = 0



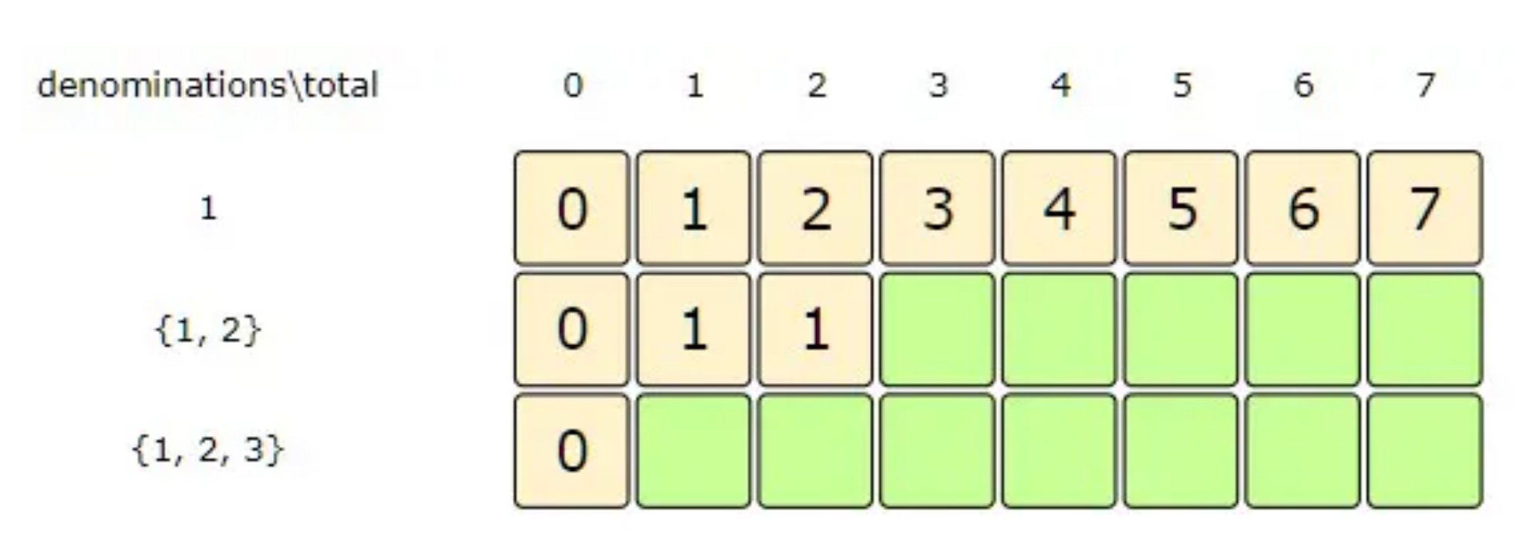
denominations\total	0	1	2	3	4	5	6	7
1	0	1	2	3	4	5	6	7
{1, 2}	0							
{1, 2, 3}	0							

Total:3-7, Index:0 => dp[Index][Total-denominations[Index] + 1, we didn't consider dp[Index-1][Total] as Index = 0



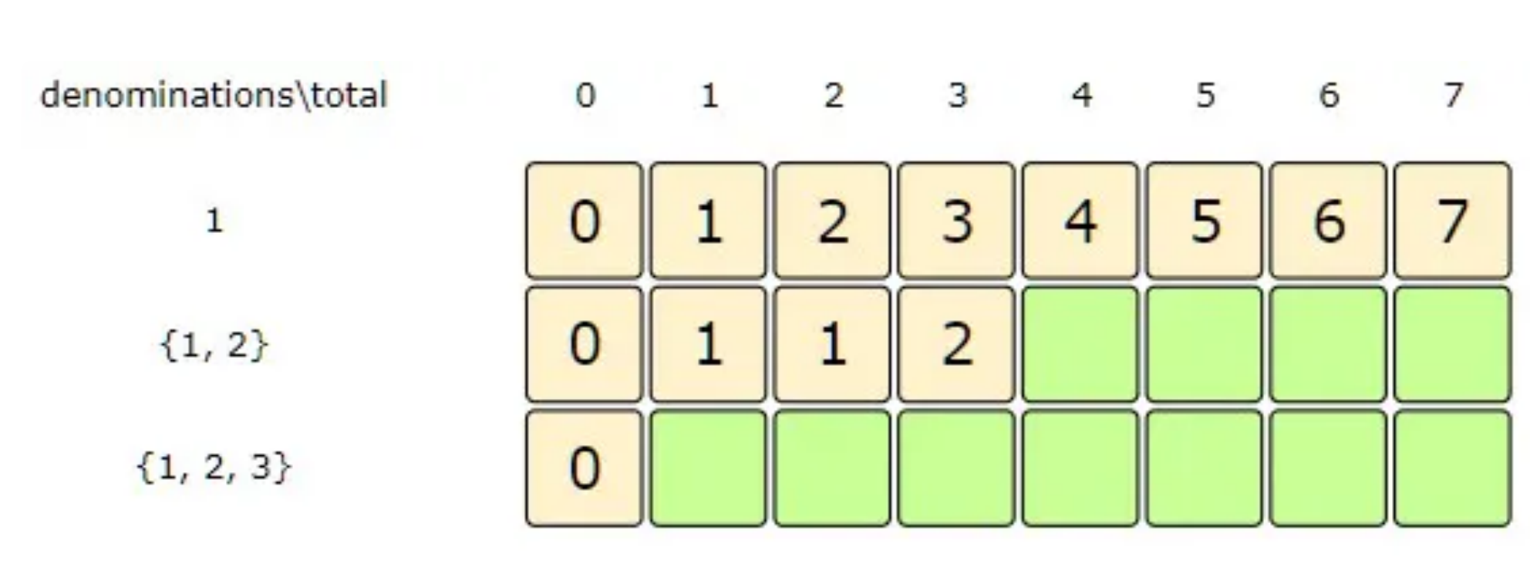
denominations\total	0	1	2	3	4	5	6	7
1	0	1	2	3	4	5	6	7
{1, 2}	0	1						
{1, 2, 3}	0							

Total:1, Index:1 => dp[Index-1][t], we didn't consider dp[Index][Total-denominations[Index] as Total < denominations[Index]



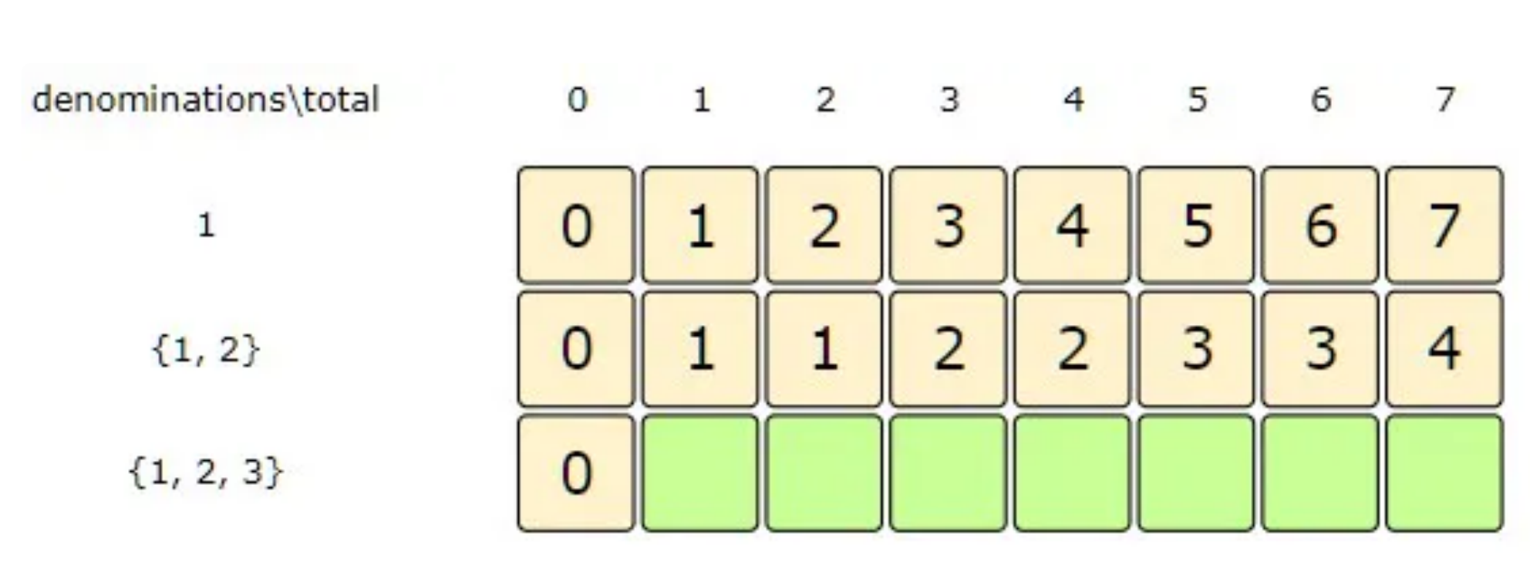
denominations\total	0	1	2	3	4	5	6	7
1	0	1	2	3	4	5	6	7
{1, 2}	0	1	1					
{1, 2, 3}	0							

Total:2, Index:1 => min(dp[Index-1][Total], dp[Index][Total-denominations[Index] + 1)



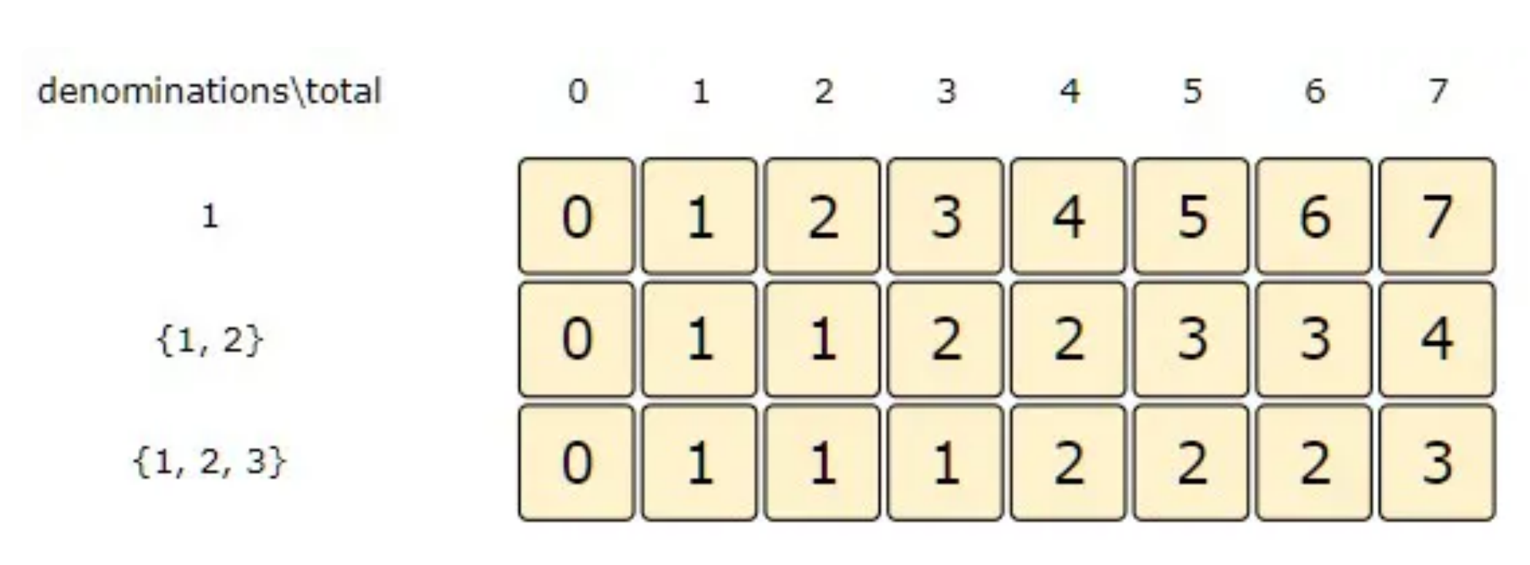
denominations\total	0	1	2	3	4	5	6	7
1	0	1	2	3	4	5	6	7
{1, 2}	0	1	1	2				
{1, 2, 3}	0							

Total:3, Index:1 => min(dp[Index-1][Total], dp[Index][Total-denominations[Index] + 1)



denominations\total	0	1	2	3	4	5	6	7
1	0	1	2	3	4	5	6	7
{1, 2}	0	1	1	2	2	3	3	4
{1, 2, 3}	0							

Total:4-7, Index:1 => min(dp[Index-1][Total], dp[Index][Total-denominations[Index] + 1)

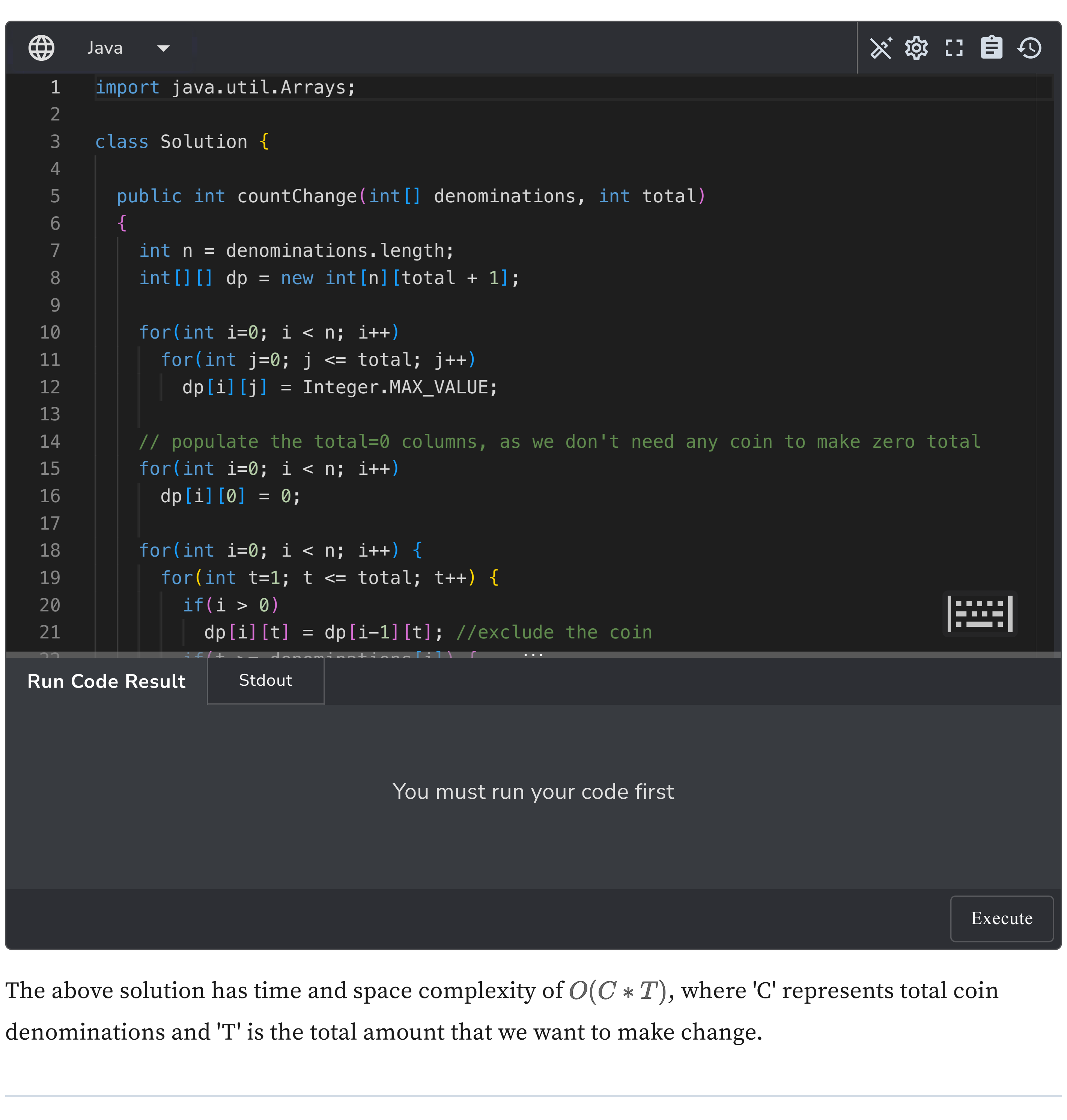


denominations\total	0	1	2	3	4	5	6	7
1	0	1	2	3	4	5	6	7
{1, 2}	0	1	1	2	2	3	3	4
{1, 2, 3}	0	1	1	1	2	2	2	3

Total:7, Index:2 => min(dp[Index-1][Total], dp[Index][Total-denominations[Index] + 1)

Code

Here is the code for our bottom-up dynamic programming approach:



```java
import java.util.Arrays;

class Solution {

  public int countChange(int[] denominations, int total)
  {
    int n = denominations.length;
    int[][] dp = new int[n][total + 1];

    for(int i=0; i < n; i++)
      for(int j=0; j <= total; j++)
        dp[i][j] = Integer.MAX_VALUE;

    // populate the total=0 columns, as we don't need any coin to make zero total
    for(int i=0; i < n; i++)
      dp[i][0] = 0;

    for(int i=0; i < n; i++) {
      for(int t=1; t <= total; t++) {
        if(i > 0)
          dp[i][t] = dp[i-1][t]; //exclude the coin
```

You must run your code first

The above solution has time and space complexity of $O(C*T)$, where 'C' represents total coin denominations and 'T' is the total amount that we want to make change.

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Minimum Coin Change

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